

SOL AutoML Executive Performance Report

Dataset: cleaned_fixed_salary_dataset (3).csv | Date: 2026-05-24 19:36:13

1. Dataset Profiling & Overview

Intelligent dataset schema identification and target analysis completed successfully.

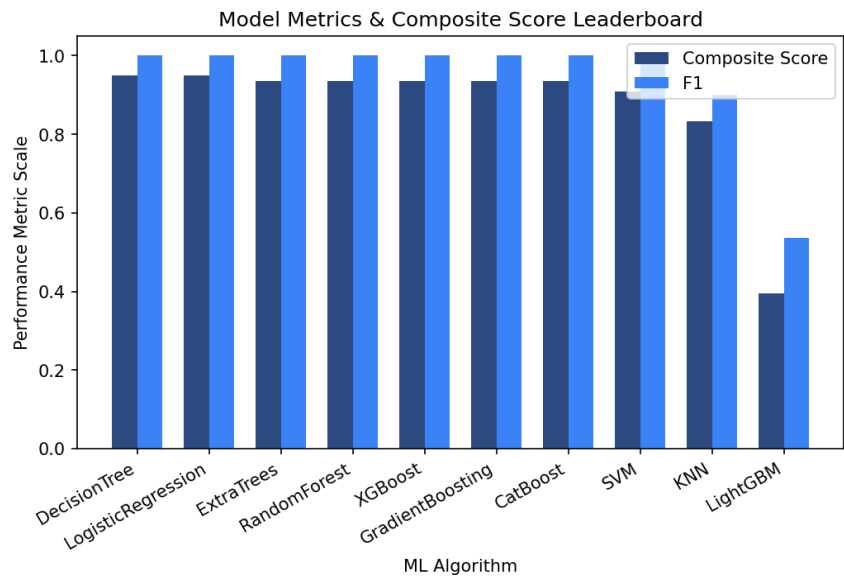
Total Dataset Rows	100
Total Dataset Columns	5
Selected Target Column	'Salary'
Inferred ML Task	Multiclass
Numerical Features	2
Categorical Features	3

2. Comprehensive Model Leaderboard & Health Evaluation

Sorted by SOL Composite Score. Overfitting, stability, variance, and complexity are fully integrated.

Model	Val F1-Score	CV Mean	CV Std	Gen Gap	Composite	Health Status
DecisionTree	1.0000	1.0000	0.0000	0.0000	0.9500	Stable
LogisticRegression	1.0000	1.0000	0.0000	0.0000	0.9500	Stable
ExtraTrees	1.0000	1.0000	0.0000	0.0000	0.9350	Stable
RandomForest	1.0000	1.0000	0.0000	0.0000	0.9350	Stable
XGBoost	1.0000	1.0000	0.0000	0.0000	0.9350	Stable
GradientBoosting	1.0000	1.0000	0.0000	0.0000	0.9350	Stable
CatBoost	1.0000	1.0000	0.0000	0.0000	0.9350	Stable
SVM	1.0000	0.9869	0.0261	0.0000	0.9087	Stable
KNN	0.8998	0.8879	0.0769	0.0621	0.8321	Slight Overfitting
LightGBM	0.5367	0.1826	0.0475	0.1101	0.3952	Slight Overfitting

Model Leaderboard Visual Comparison

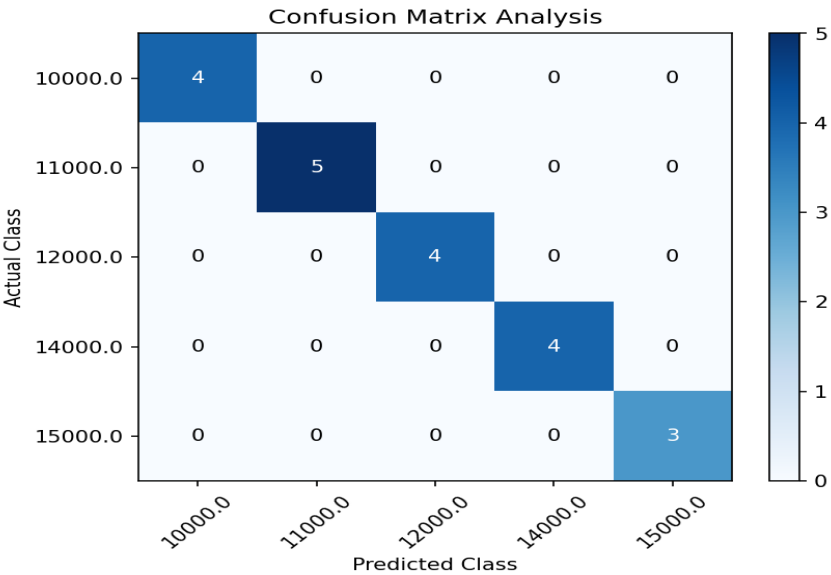


3. Champion Model Deep Performance Analysis

The selected champion algorithm is: DecisionTree.

Performance Evaluation Metric	Train Set	Validation Set
ACCURACY	1.00000	1.00000
PRECISION	1.00000	1.00000
RECALL	1.00000	1.00000
F1	1.00000	1.00000
ROC AUC	1.00000	1.00000

Confusion Matrix Analysis



4. Relative Feature Influence Profiles

Feature Column	Relative Influence Impact
Job_Title_Mobile Developer	31.61%
Job_Title_Frontend Developer	25.38%
Job_Title_Backend Developer	22.59%
Job_Title_DevOps Engineer	20.42%
Age	0.00%
Name_Ahmed Ali	0.00%
Name_Ahmed Fawzy	0.00%
Name_Ahmed Hassan	0.00%

5. Production Deployment Instructions

- Deploy 'trained_model.pkl' and 'preprocessing_pipeline.pkl' into your production environment.

2. Use 'predict.py' as your standalone deployment framework for zero-configuration setup.
3. Preprocess inference payloads strictly through the loaded preprocessor before feed predictions.
4. Setup shadow deployments and shadow tests initially to verify stability under heavy workloads.